ANDREW NEIL

AI ENGINEERING | AGENTIC AUTOMATION | PYTHON BACKEND

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PROFESSIONAL SUMMARY

Computer Engineering student (2027) with hands-on experience building Python automation, LLM-integrated desktop agents, Django backend workflows, computer-vision prototypes, and quantitative trading research tools. Combines practical client delivery with project work in tool-using AI systems, testing, APIs, and hardware-integrated applications.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL, C/C++
AI & automation: LLM APIs (Gemini, OpenAI, Claude), local LLMs, agent/tool calling, LangChain, prompt workflows, Playwright, Selenium, PyAutoGUI
ML & computer vision: Pandas, NumPy, scikit-learn, PyTorch, OpenCV, YOLOv8, OCR, pandas-ta
Backend & data: Django, FastAPI, REST APIs, relational databases, Parquet, SMTP, pytest
Tools & systems: Git/GitHub, Linux/CLI, Flutter, React, Tailwind CSS, ESP32, RFID, JWT

EXPERIENCE

Backend Engineer Intern | *Aug 2026 – Present*

41studio

Tech: Python, Django, relational databases

- Contributed to a Django-based ERP for a polytechnic college, supporting student records and academic administration across department heads, teachers, classes, and students.
- Worked on backend workflows for marks entry, attendance tracking, student-data management, marksheet generation, and certificate generation.
- Helped connect role-specific workflows into a centralized academic system.

Freelance Python Developer (Selected Upwork Contracts) | *2023 – 2025*

Independent client projects

Tech: Python, Selenium, LangChain, OpenAI API, Claude API, SMTP

- Built a website-monitoring script that checked appointment availability and emailed alerts using the client-provided SMTP server and port.
- Created a command-line LLM workflow that generated three response variants, supported regeneration and user selection, and reused selected outputs as context; used LangChain with OpenAI/Claude APIs.
- Delivered Selenium-based browser automation and web-scraping work; completed three fixed-price contracts and received two five-star client reviews.

EDUCATION

B.E. in Computer Engineering | Xavier Institute of Engineering (XIE), Mumbai | Expected 2027

CGPA: 7.48/10 (through Semester VI) | Relevant coursework: Data Structures, DBMS, Operating Systems, Computer Networks, AI/ML, IoT

SELECTED PROJECTS

OSCAR — Multi-Agent AI Desktop Automation Assistant

Personal project

Tech: Python, Flutter, Gemini, local LLMs, OmniRoute/FreeLLMAPI, Playwright, PyAutoGUI, YOLOv8, OCR, pytest

- Built a Windows-focused agentic assistant that converts natural-language tasks into workflows across browser automation, desktop interactions, and terminal tasks, using a Python orchestration backend and Flutter desktop UI.
- Implemented screenshot-driven planning and JSON action execution, with browser/UI/terminal agents, persistent workflow state, and tool-payload validation; added record-and-replay for mouse, keyboard, delays, and coordinates.
- Added reliability safeguards including fresh observations, bounded retries, outcome verification, duplicate-typing prevention, and explicit completion checks; 336 automated regression tests pass using mocked Windows APIs.

Repository: github.com/smmsa1912/OSCAR

TradeAssist — Quantitative Trading Research & Risk Management System

Personal project

Tech: Python, Angel One SmartAPI, Pandas, NumPy, pandas-ta, Parquet, scikit-learn, PyTorch

- Built a historical market-data pipeline and backtesting engine for Indian equities, covering technical indicators, entry/exit signals, historical trade simulation, brokerage costs, slippage, stop-losses, and trailing stops.
- Implemented risk controls and portfolio analysis including position sizing, exposure limits, P&L, equity curves, maximum drawdown, Sharpe ratio, win rate, profit factor, buy-and-hold comparison, and walk-forward/out-of-sample evaluation.
- Best recorded IRFC SMA 20/50 crossover with 5% stop-loss backtest (Jan 2023–Sep 2026, 13 walk-forward folds): +24.17% cumulative return, 6.75% CAGR, −11.03% max drawdown across 86 trades, with 5-bps slippage and modeled costs. Backtest results only; no live-profitability claim.

FinVerse — AI-Powered Blockchain Financial Operating System

Final-year project

Tech: React, Tailwind CSS, Django, Flutter, Python, blockchain, ESP32, NFC/RFID

- Built an integrated financial-platform prototype combining a web dashboard, Django backend, mobile application, blockchain components, AI-assisted financial workflows, and ESP32/NFC hardware authentication.
- Connected financial and authentication components into an end-to-end project prototype; not represented as a deployed commercial product.

Dynamic Toll Charging System

College mini-project; competition winner

Tech: Python, YOLOv8, OpenCV, OCR/ANPR, RFID

- Developed a tollgate demonstration combining vehicle detection, number-plate recognition, RFID-based vehicle identification, and a dummy wallet; RFID simulated FASTag and payment processing was not implemented.
- Won the college mini-project competition; co-authored a 2026 IJIRT paper on the system.

KYC Smart Card — RFID-Based Digital Document Platform

Academic mini-project

Tech: Django, Flutter, ESP32, RFID, Bluetooth LE, JWT, document encryption

- Co-developed a DigiLocker-inspired platform for storing and sharing documents with institutions, combining a Django web backend, RFID smartcard verification, and mobile/Bluetooth authentication.
- Published project research in IJIRT describing the document-management and dual-mode authentication design; website backend contribution is identified in the earlier paper.

PUBLICATIONS

Neil, A. et al. “Dynamic Toll Charging System.” *IJIRT*, Vol. 12, Issue 8, 2026, pp. 5716–5719. Read paper

Neil, A. et al. “KYC Smart Card” (expanded dual-mode authentication version). *IJIRT*, Vol. 12, Issue 1, June 2025. Read paper (PDF)

Neil, A. et al. “KYC Smart Card” (initial RFID document-repository version). *IJIRT*, Vol. 11, Issue 7, December 2024. Read paper (PDF)